

EMG-CONTROLLED LOW-COST PROSTHETIC ARM

MINI PROJECT FINAL REPORT

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**MAR BASELIOS COLLEGE OF ENGINEERING AND TECHNOLOGY
(Autonomous)**

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APRIL 2026

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*Submitted in partial fulfilment of the requirements of the award of
Bachelor of Technology Degree in Electrical and Computer
Engineering of the APJ Abdul Kalam Technological University (KTU)*

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CERTIFICATE

*This is to certify that this report entitled “**EMG-CONTROLLED LOW-COST PROSTHETIC ARM**” is a bonafide record of the seminar presented by **ABRAHAM JACOB (MBT23EL005, B23EE2105), ANNA SUSAN GEORGE (MBT23EL005, B23EE2105), HANOK SAMUEL BOBAN (MBT23EL005, B23EE2131) AND NAVIN TOM BABU (MBT23EL005, B23EE2105)** towards the partial fulfilment of the requirements for the award of B.Tech Degree in Electrical and Computer Engineering of the APJ Abdul Kalam Technological University (KTU) during the year 2026.*

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ABSTRACT

This mini project presents a low-cost prosthetic arm designed to address the high cost and mechanical complexity of conventional prosthetic systems. The proposed design is inspired by the gripping mechanism of an octopus tentacle and utilizes a soft, adaptive gripper for improved functionality. The system operates using only two motors: one to achieve tentacle-like gripping through a single-string mechanism, and another to control an integrated suction cup for handling flat or smooth objects. A single EMG signal is employed to activate the gripping action, thereby simplifying the control system and reducing unintended movements. In addition, the project explores a complementary software approach for user-specific customization, wherein input measurements of the residual limb can be used to generate a 3D model of the prosthetic socket suitable for 3D printing. By minimizing components and adopting a bio-inspired approach, the design enhances affordability, ease of maintenance, and usability, making it a practical and scalable alternative to existing prosthetic solutions, particularly in resource-constrained settings.

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NOMENCLATURE

- EMG : Electromyography
- DC : Direct Current
- UI : User Interface
- PWM : Pulse Width Modulation
- Li-ion : Lithium-ion battery

1. INTRODUCTION

1.1. BACKGROUND AND CONTEXT

The loss of an upper limb can have a huge effect on a person's quality of life. It can fundamentally alter the way they carry out their routine activities and perhaps even influence their emotional and psychological well-being in ways that are not always easy to measure. Prosthetic devices have long been the primary means of helping restore some of the lost functionality. Over the decades, extensive research and engineering efforts have been undertaken to make these devices advanced, capable, and responsive to the needs of those who rely on them.

However, the current landscape of upper-limb prosthetics remains deeply divided. On one end are highly advanced, multi-articulated bionic arms that provide impressive levels of dexterity, but come with extremely high price tags often running into several lakhs of rupees as well as significant maintenance demands that make them largely impractical outside well-equipped clinical environments. On the other end are simple, body-powered hooks, which are much more affordable, but fall short when it comes to both appearance and the level of functional support they can offer in everyday life.

The integration of EMG signals into prosthetic control marked an enormous leap forward. By using EMG signals, amputees are able to control prosthetic limbs through the remaining electrical impulses in their stump muscle tissues, a method that is far more intuitive than relying on external controls such as switches or levers. Yet, despite this advancement, most existing EMG-controlled artificial arms still employ several motors, usually one for each finger, along with complex tendon and pulley systems to enable movement. These mechanisms present challenges not only because of their high initial cost, but also due to the ongoing maintenance they require.

1.2. THE CORE CHALLENGE

Conventional prosthetic arms are fundamentally expensive and mechanically fragile. The inclusion of multiple small actuators and complex internal mechanisms increases both manufacturing cost and the probability of mechanical failure over time. In low-resource settings whether in rural India or in developing economies around the world

replacement parts and trained repair technicians are simply not available. This means that when these devices break down, users are left without their assistive technology indefinitely.

Beyond mechanical fragility, there is a serious human-factors problem. Multi-channel EMG control requires users to learn to isolate and flex specific remnant muscles to trigger specific finger movements. This is a cognitively demanding process that many users find exhausting and unreliable. Factors as simple as sweat, muscle fatigue, or a slightly shifted socket fit are enough to degrade the signal quality and cause the arm to behave unpredictably. High abandonment rates for advanced prosthetics are a well-documented outcome of this problem.

Ultimately, advanced prosthetics remain inaccessible to the vast majority of amputees globally, especially in developing countries where access to medical facilities and experts is not readily available. This leads to the deprivation of autonomy and means of earning a living for many individuals.

1.3. THE PROPOSED INNOVATION

To bridge the gap between affordability and meaningful functionality, this project takes a fundamentally different design approach—one rooted in biomimicry. Rather than engineering the prosthetic arm to look and behave like a human hand, we asked the question: what biological system is exceptionally good at grasping objects of many shapes and sizes, using a minimum of complex parts?

The answer, which forms the creative foundation of this project, is the octopus tentacle. An octopus does not grip objects using rigid, individually actuated fingers. Instead, it wraps a continuous, flexible structure around an object, conforming to its shape and distributing grip force naturally. This same principle is what drives the mechanical design of our proposed prosthetic arm.

Our design replaces the traditional five-finger array with a single, unified soft gripper structure actuated by just two motors. A primary continuous servo motor handles the wrapping and gripping motion through a simplified single-string tendon, while a secondary DC motor powers a suction mechanism for grasping objects that a mechanical gripper might struggle with such as plates, glass surfaces, or smartphone

screens. Crucially, the entire control scheme runs on a single-channel EMG signal, drastically reducing the learning curve compared to conventional multi-channel systems.

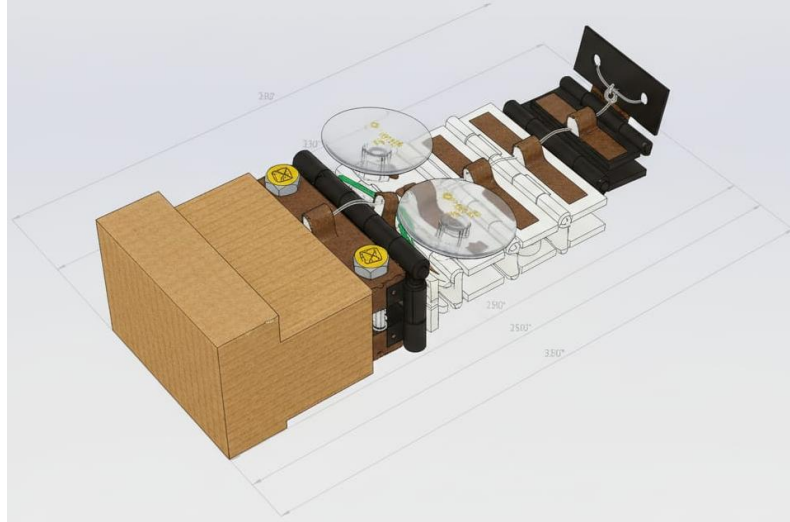


Figure 1.1 Representation of proposed prosthetic arm

1.4. SUMMARY

Losing an upper limb could have a significant impact on one's life, making it difficult for him/her to function independently. Even though there have been incredible developments in the field of prosthetics, advanced devices are too costly and intricate, whereas inexpensive devices do not have any functional value. This project aims to design a prosthetic arm that is relatively low-cost and simple in nature. It will be based on the flexibility demonstrated by the octopus tentacle. Instead of using fingers for gripping things, the device will have a basic flexible gripper, which will be actuated by two motors using a single-channel EMG setup.

2. LITERATURE REVIEW

2.1. INTRODUCTION

A thorough understanding of existing research in soft robotics, biomimetic grippers, and prosthetic control systems is essential for grounding the innovations of this project in the context of what has come before. The following three works were carefully studied to identify the gaps in current technology and to justify the specific architectural choices made in this design.

2.2. MULTIFUNCTIONAL SOFT GRIPPERS IN ROBOTICS

Shin, J., Kwon, J., "Multifunctional Soft Gripper with Various Grip Modes via Single Linear Actuation and One-Shot Fabrication," IEEE Access, vol. 13, 2025.

This paper presents a genuinely innovative soft robotic gripper that achieves multiple distinct grip modes using only a single linear actuator. The authors demonstrate, convincingly, that mechanical complexity at the output can be achieved through clever material geometry and structural design rather than through multiple actuators a concept philosophically aligned with our own work. Their fabrication methodology, which they describe as "one-shot," produces a soft gripper capable of adapting to objects of different shapes from a single continuous input force.

However, the limitations of this design become clear when evaluated against prosthetic use requirements. The gripper is meant to be used for manipulation purposes in robotics. This design makes use of materials that have been specially formulated and manufactured, and cannot withstand the physical stress and wear and tear that a prosthetic limb will undergo daily. More critically, the system has no EMG interface, no consideration for wearability or socket design, and does not address the load-bearing demands of practical human tasks.

Our project adopts the philosophy of this work single-actuation simplicity yielding rich mechanical behaviour but applies it through a mechanically robust, octopus-inspired gripping structure specifically designed for human prosthetic use. We pair

this with active suction for edge-case grasping and a user-friendly EMG control layer, making it a solution genuinely suited to the demands of daily life.

2.3. BIOMIMETIC CONTINUUM ROBOTICS

Ma, R., Peng, J., Yu, P., Zhao, N., "Biomimetic design and kinematic analysis of a rope-driven soft robotic arm," IEEE CACRE, 2024.

This work explores the design and kinematic modelling of a rope-driven soft robotic arm inspired by biological tentacle structures. The authors develop detailed mathematical models for trajectory planning and demonstrate the system's ability to navigate unstructured environments with considerable flexibility. The rope-driven (or tendon-driven) methodology described in this paper is directly relevant to our own single-string tendon mechanism.

Yet despite these technical strengths, the system described by Ma et al. relies on multiple servo motors to achieve its various degrees of freedom, and the control architecture requires complex computational models to generate even simple trajectories. This makes the system entirely unsuitable for a low-cost, wearable prosthesis where the entire controller must fit within a small socket, run on a battery pack, and be maintained by a non-specialist user. The weight, cost, and fabrication complexity of a multi-servo rope-driven arm disqualifies it from the target use case.

Our design preserves the elegance of the rope-driven gripping concept and removes the complexity by using a single continuous servo spools a single string, which in turn actuates the entire gripper as a unified structure. This results in a system that is lightweight, power-efficient, and crucially easy to repair when something goes wrong.

2.4. ANTHROPOMORPHIC TRANSHUMERAL PROSTHESES

Lenzi, T., Lipsey, J., Sensinger, J. W., "The RIC Arm — A Small Anthropomorphic Transhumeral Prosthesis," IEEE/ASME Transactions on Mechatronics, vol. 21, no. 6, pp. 2660–2671, Dec. 2016.

The RIC (Rehabilitation Institute of Chicago) Arm represents the high end of what conventional prosthetic engineering can achieve. It is designed to closely replicate the kinematic behaviour of the human arm through an array of actuators, joints, and an advanced multi-channel EMG control system. Functionally, it is impressive capable of performing tasks that simpler prosthetics cannot.

However, the design elements that contribute to the functional ability of the RIC Arm are also responsible for its high cost, weight, and maintenance requirements. The dense packing of actuators and joints increases overall weight, leading to socket discomfort during extended use. The complexity of the internal mechanisms means that a single mechanical failure often requires the entire unit to be professionally serviced an option simply not available to the majority of amputees in low-resource environments.

Our project takes a deliberately opposing philosophy. By reducing the component count to the absolute minimum needed for effective daily-use grasping, we achieve a design where part replacement is fast, cheap, and potentially manageable by the user themselves. The absence of complex gear trains and finger joints is not a limitation it is a deliberate engineering choice that directly serves the target user population.

2.5. SUMMARY

The literature examined suggests that despite the remarkable functionality of these systems, there is always a problem with the complexities involved, which tend to make such devices costly and hard to maintain. In other words, soft grippers exemplify the advantages of simplicity of actuation, tendon-driven continuum robots emphasize flexibility, and sophisticated prosthetic arms exhibit an exceptional degree of controllability and mobility. Building upon these findings, this project attempts to incorporate the strengths of these studies into the development of a prosthetic arm that will be cheap, durable, and usable.

3. PROBLEM STATEMENT AND OBJECTIVES

3.1. PROBLEM STATEMENT

The current state of upper-limb prosthetics presents a painful contradiction. The technology exists to create arms with impressive dexterity and control but the people who need prosthetics most are the least able to access these advanced solutions. In the developing world and in lower socio-economic settings globally, amputees are routinely forced to choose between two inadequate options: an affordable but functionally limited body-powered hook, or a highly capable but prohibitively expensive and fragile bionic hand.

Conventional advanced prosthetic arms are mechanically complex by design. They rely on five or more individual motors to drive individual fingers, with intricate tendon and gear mechanisms linking actuator motion to finger movement. This structural density inflates the manufacturing cost well beyond ₹1,00,000 for even entry-level commercial devices. When these complex systems fail and they do fail users are left without their assistive technology for extended periods while awaiting specialized repair.

The control paradigm compounds the problem. Multi-channel EMG control systems require the user to isolate and flex specific remnant muscles to trigger specific hand gestures. This process is very demanding, highly susceptible to variables such as muscle exhaustion, sweating, and socket displacement, and produces highly unreliable outcomes. The expense involved, poor reparability, and difficult learning curve have resulted in many abandoned units, even for those that have successfully been fitted with sophisticated prosthetics.

3.2. PROPOSED OBJECTIVES

- **Biomimetic Redesign:** To design a gripping mechanism inspired by the octopus tentacle, replacing the conventional five-finger array with a single, robust, adaptive gripper that conforms to object geometry naturally.

- **Mechanical Simplification:** To drastically reduce mechanical complexity by operating the entire prosthetic arm using only two motors — one continuous servo for the mechanical gripping motion, and one DC motor for the suction mechanism.
- **Intuitive Control:** To simplify the human-machine interface by utilizing a single-channel EMG signal, ensuring the system is easy to learn, works reliably in daily conditions, and does not require the user to perform complex muscle isolation exercises
- **Enhanced Adaptability:** To improve the prosthesis's ability to grasp a wide variety of objects varying in shape, size, and surface texture by combining the tentacle-like wrapping grip with an auxiliary suction cup for smooth or flat objects.
- **Cost and Maintenance Reduction:** To architect a modular, off-the-shelf system where components are easily sourced and individually replaceable, targeting a total build cost in the ₹12,000 – ₹15,000 range representing approximately a 90% reduction compared to conventional devices.
- **Software Interface:** To develop a dedicated mobile application that allows users to calibrate the system to their personal muscle signal profile, adjust sensitivity, and select preset grip modes suited to common daily tasks.

3.3. SUMMARY

The current scenario of upper limb prostheses often leaves the user with a choice between cost-effectiveness and efficiency. High-end bionic limbs provide good functionality but are costly, complicated, and hard to maintain, whereas cheaper variants often lack practicality. The objective of this study is to fill that void by creating a cost-effective and efficient prosthetic arm that incorporates the adaptable grasping capabilities of an octopus tentacle. The device will be built using a dual motor arrangement, a single-channel EMG sensor, and a suction mechanism. It will offer intuitive control, multi-purpose object manipulation, easy maintenance, and high cost-effectiveness. A mobile application will enhance usability by enabling customization and switching between various gripping modes.

4. SYSTEM ARCHITECTURE

4.1. INTRODUCTION

EMG-Controlled Low-Cost Prosthetic Arm is arranged in a modular manner using two distinctly identifiable layers. These include the hardware layer that performs physical functions such as actuation and signal acquisition, and the software layer that takes charge of managing logical functionality of the system and customizations done by users. The modularity of the design allows each layer to function independently while being part of a whole system. Modularity was introduced as an intentional strategy meant to make debugging easier, minimize complexities of the design and facilitate future system improvement without changing both layers at once.

4.2. SYSTEM BLOCK DIAGRAM

The ESP32 microcontroller sits at the centre of the entire system, acting as the computational hub. It continuously receives analog input from the EMG sensor, processes the signal for muscle activity, and converts the information into PWM control signal to drive the L298N motor driver. The motor driver, in turn, manages the power delivery to both the continuous servo motor and the suction DC motor.

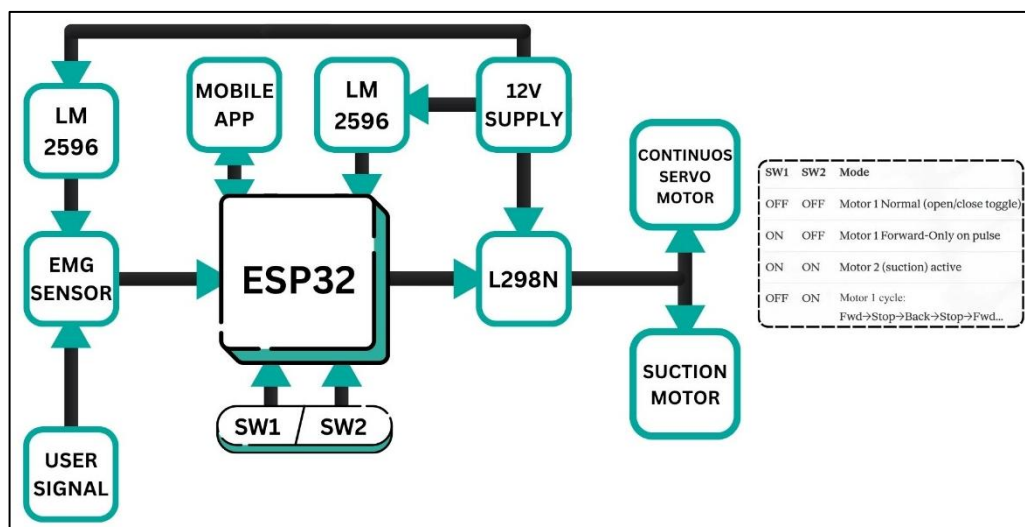


Figure 4.1 Block diagram of the prosthetic arm system

As shown in Figure 4.1, the entire system draws power from a central 3S Lithium-ion battery pack operating at a nominal 11.1V. The raw battery voltage is stepped down by two independent LM2596 buck converters, providing clean regulated voltages to the ESP32 and the EMG sensor. The motor driver receives the full battery voltage directly to handle the higher current demands of the actuators. Two physical tactile switches (SW1 and SW2) are connected directly to the ESP32, thus enabling the user to switch between four modes of operation without having to use a phone. The mobile application communicates with the ESP32 over Bluetooth for calibration and advanced control.

4.3. CORE HARDWARE COMPONENTS

4.3.1. Actuators: Continuous Servo and Suction Motor

- **Continuous Servo Motor (150Kg torque):** This forms the mechanical core of the gripping system. The main difference between a regular positional servo and a continuous rotation servo lies in the fact that the latter can rotate infinitely in both directions and operate at various speeds. This capability is crucial for the single-string tendon system: when the servo starts turning, the string will wind up, causing the gripper to wrap around and clamp down on the object. Once the servo stops working, the string will unravel, and the gripper will be released.
- **Suction Motor and Suction Cup:** A motor from 6 to 12V DC powers a small vacuum pump attached to a silicone suction cup affixed to the gripper. This provides a secondary gripping mode specifically for objects that a mechanical gripper struggles with such as flat plates, glass screens, or smooth-surfaced items. The suction mechanism can be activated independently through the app or via EMG in Mode 3, allowing the user to switch strategies based on what they are trying to pick up.

4.3.2. Sensors and Control Logic

- **EMG Sensor (Muscle BioAmp Patchy or equivalent):** The EMG sensor is the primary human input interface. It adheres to the user's residual limb

and detects the micro-voltage electrical signals produced whenever the user contracts their muscles. These raw analog signals are fed into the ESP32's ADC input, where they are processed through a software envelope-detection algorithm to extract clean, reliable activation events from the inherently noisy raw signal.

- **Microcontroller (ESP32):** The choice of the ESP32 over simpler microcontrollers such as the Arduino Uno was made for a number of reasons that relate to fulfilling the requirements of this project. For instance, its two-core processor and hardware timer interfaces provide consistent and accurate PWM signal generation, even when executing Bluetooth simultaneously. More importantly, its Wi-Fi and Bluetooth capabilities enable wireless communication between the prosthetic device and the mobile calibration app without requiring additional hardware. The ESP32 runs the full control state machine including four operating modes, EMG calibration, signal thresholding, and Bluetooth command parsing in a single embedded program.

4.3.3. Power Management

- **Motor Driver (L298N):** The L298N dual H-bridge motor driver provides a safe interface between the ESP32's 3.3V logic pins and the 12V motors. It handles the high inrush current that happens during startup with load. In addition, there are flyback diodes included that will help protect the circuit from inductive spikes caused when turning off the motors. The L298N motor driver will control both Motor 1 and Motor 2, with Motor 1 being the servo motor and the gripper while Motor 2 is the suction motor..
- **Buck Converters (LM2596):** Two separate LM2596 step-down converters are used in the power chain. The division of the voltage regulation task to two discrete regulators, where one regulates the voltage to the ESP32 logic power and the second regulates the voltage to the EMG sensor, helps ensure that the switching noise from the microcontroller will not interfere with the EMG signal. This was an important feature that improved the Signal-to-Noise Ratio during practical implementation.

- **Power Supply (3S Li-ion Battery Pack, 11.1V):** Since lithium-ion batteries have an extremely high energy density, the arm will be able to operate throughout the entire day without any need for intermediate charging.

4.4. SUMMARY

This prosthetic arm has a modular construction which includes both hardware and software technologies. The ESP32 microcontroller receives EMG data and runs the gripper mechanism along with the suction mechanism by controlling the L298N motor driver. The gripper mechanism is powered by a continuous servo while the suction mechanism uses a DC motor. This arm is run on a Li-Ion battery which regulates voltage for a smooth operation. Bluetooth connectivity and physical switches allow easy control, calibration, and mode selection, making the system simple, efficient, and user-friendly.

5. WORKING OF THE GRIPPING MECHANISM

5.1. INTRODUCTION

The octopus-inspired gripper is at the centre of the prosthetic arm, as it determines its object manipulation capabilities. The prosthetic hand doesn't have numerous rigid fingers, but a flexible tendon-driven one which will adapt its form according to the object it manipulates. Consequently, the mechanism will be able to hold any item irrespective of its size or shape without posing any difficulties for its handler during the process. In addition, the mechanism will still be simple in design, ensuring that it performs effectively.

5.2. WORKING PRINCIPLE

The gripping is accomplished by employing a continuous rotation servo motor of high torque that is connected to a tendon string, making it possible for the device to utilize its spinning movement to grip objects effectively.

- When the servo rotates forward, the tendon winds around the servo horn.
- This pulls the internal cable running through the flexible gripper.
- As tension increases, the gripper curls inward, wrapping around the object.

To release the object:

- The servo rotates in the reverse direction.
- The tendon unwinds.
- The gripper returns to its open position.

This simple process of winding and unwinding eliminates the necessity for several motors, gears, and intricate linkage systems, thus resulting in an easily manageable system that is mechanically less complex and more reliable in the long run.

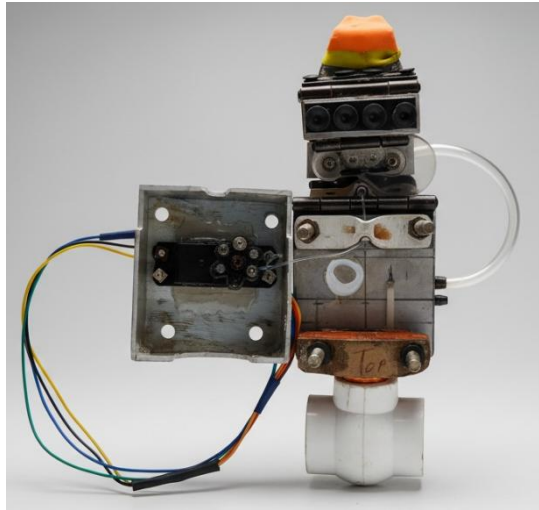


Figure 5.1 Tendon-driven gripper mechanism used in the prosthetic arm

5.3. ADAPTIVE OBJECT GRASPING

One of the key strengths of the system proposed is that it is able to self-regulate depending on the size of the object to be processed; this means that the system is able to handle any number of objects regardless of shape.

- The cylindrical items like bottles and cups are tightly packed in order to have good and tight grip.
- Odd-shaped objects get contoured according to the gripper's curved surface, thereby making them easy to handle.
- A dedicated pinching mechanism, integrated as an additional structural feature, enables precise handling of small objects
- The tentacle is capable of forming a closed ring shape around items like rolls of tape, handles, ropes, or rings, which makes for easy picking up and handling of these items.

This adaptation enhances the effectiveness of the grip system while reducing the complexity of control required by the operator, making it practical and easy to operate under normal conditions.

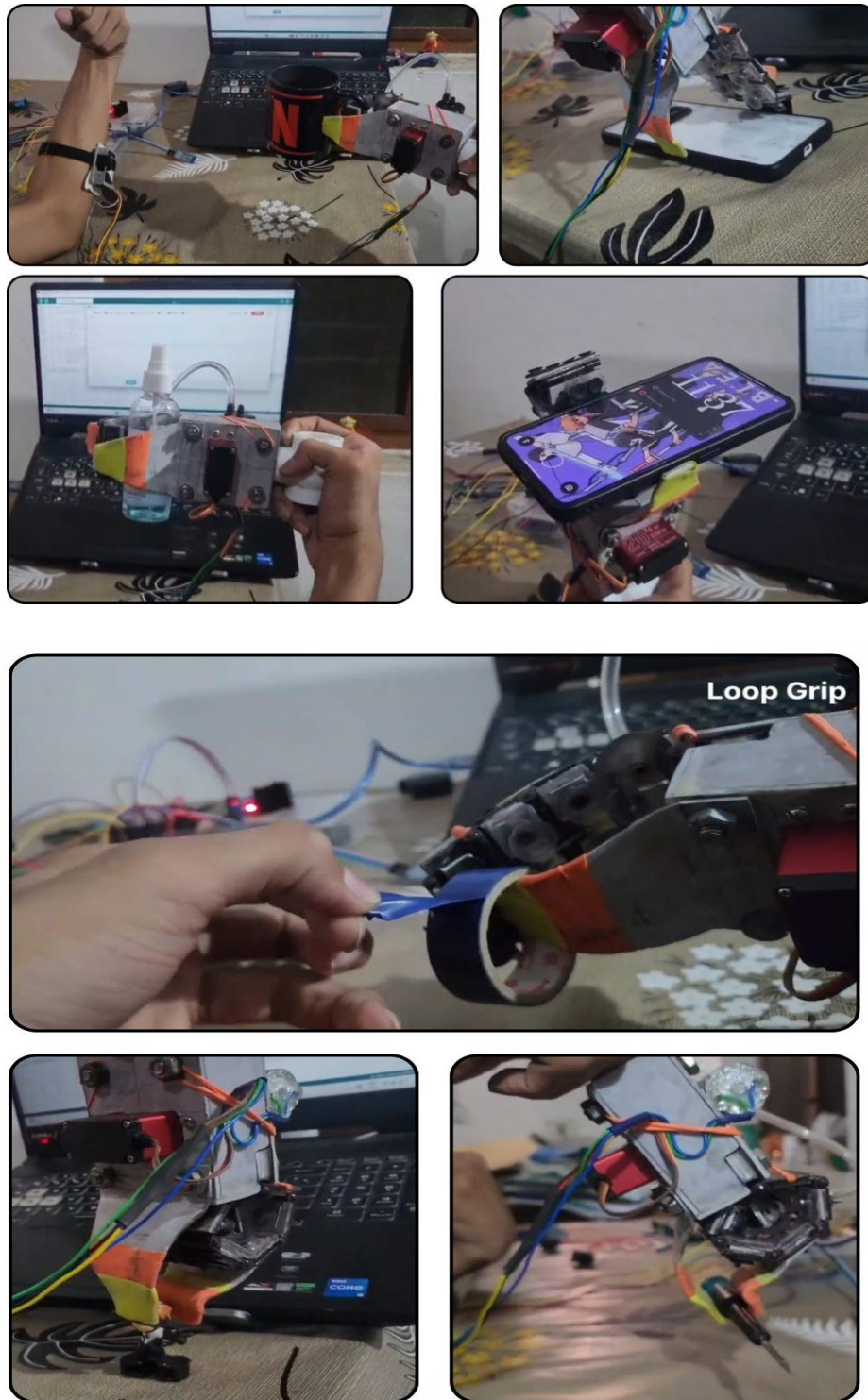


Figure 5.2 Demonstration of adaptive object grasping using the prosthetic arm on objects of varying shapes and sizes.

5.4. SUCTION-ASSISTED GRIPPING

For objects that are difficult to grasp mechanically, such as glass sheets, smartphones, or flat containers, the integrated suction system provides additional holding capability.

- When activated, the DC vacuum pump creates negative pressure.
- The object adheres firmly to the suction cup.

This hybrid gripping approach greatly expands the range of objects the prosthetic can manipulate.

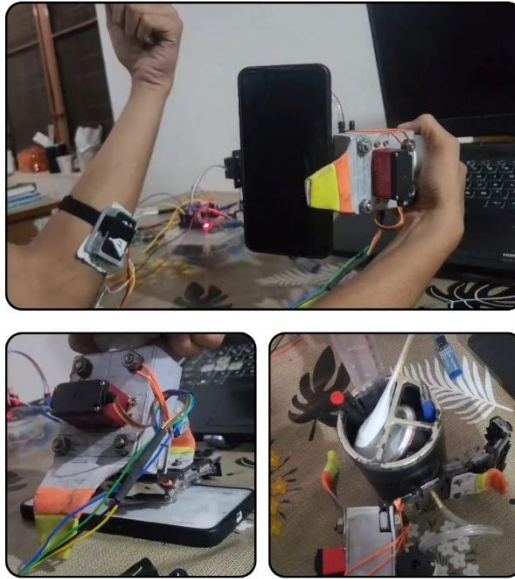


Figure 5.3 Demonstration of suction-assisted gripping for handling flat and smooth objects using the prosthetic arm.

5.5. SUMMARY

The octopus-inspired gripping mechanism is the core innovation of this prosthetic arm, offering a simple yet highly adaptable alternative to traditional multi-finger designs. The high-torque continuous servo acts on a single tendon to wind and unwind it, which leads to the curling and uncurling of the flexible gripper around the object. The tendon-based design allows for the absence of complicated gears and several motors in its construction while giving the gripper the ability to adjust to different object geometries naturally. The system is able to hold onto cylindrical and non-cylindrical objects, manipulate small objects accurately, and wrap around larger objects, like handles and tape reels, in a closed-loop fashion.

6. SOFTWARE AND USER INTERFACE

6.1. INTRODUCTION

The physical hardware of a prosthetic arm is only as effective as the system that governs it. A sensor and two motors mean nothing if the control is poorly calibrated or impossible to adapt. The software ecosystem of this project, a custom mobile application communicating with the ESP32 over Bluetooth, and a carefully designed embedded control program, addresses this directly.

6.2. SIGNIFICANCE OF SOFTWARE CALIBRATION

Human physiology is highly individualistic and varies from person to person. The size of the EMG signal can vary considerably depending on who is wearing the prosthetic device, even if they perform the same muscle movement. Additionally, the nature of the EMG signal for the same person can vary during different hours of the day because of variables like muscle fatigue, sweat, slight variations in the placement of the electrodes, and fit of the socket within the residual limb.

As a result, implementing a control mechanism based on hardcoded thresholds would not produce accurate results. The threshold levels would prove to be too sensitive in some cases leading to inappropriate activation. In contrast, the thresholds which would suit some people might fail to work for others. Moreover, even for a certain individual, performance can vary through the course of the day.

For this reason, the microcontroller relies on software-based calibration. During the startup phase or whenever prompted by the wearer, the ESP32 records hundreds of EMG signals from resting muscles. From these signals, a baseline is established, and the device uses this value to set its activation threshold.

6.3. MOBILE APPLICATION FEATURES

The mobile app called TentaKai, created using Flutter for multi-platform functionality, connects to the ESP32 through its internal Bluetooth chip. It works as both a configurator, monitor, and instruction manual at once. The application was

designed with accessibility in mind and includes features such as high contrast and large touch controls optimized for single-handed usage.

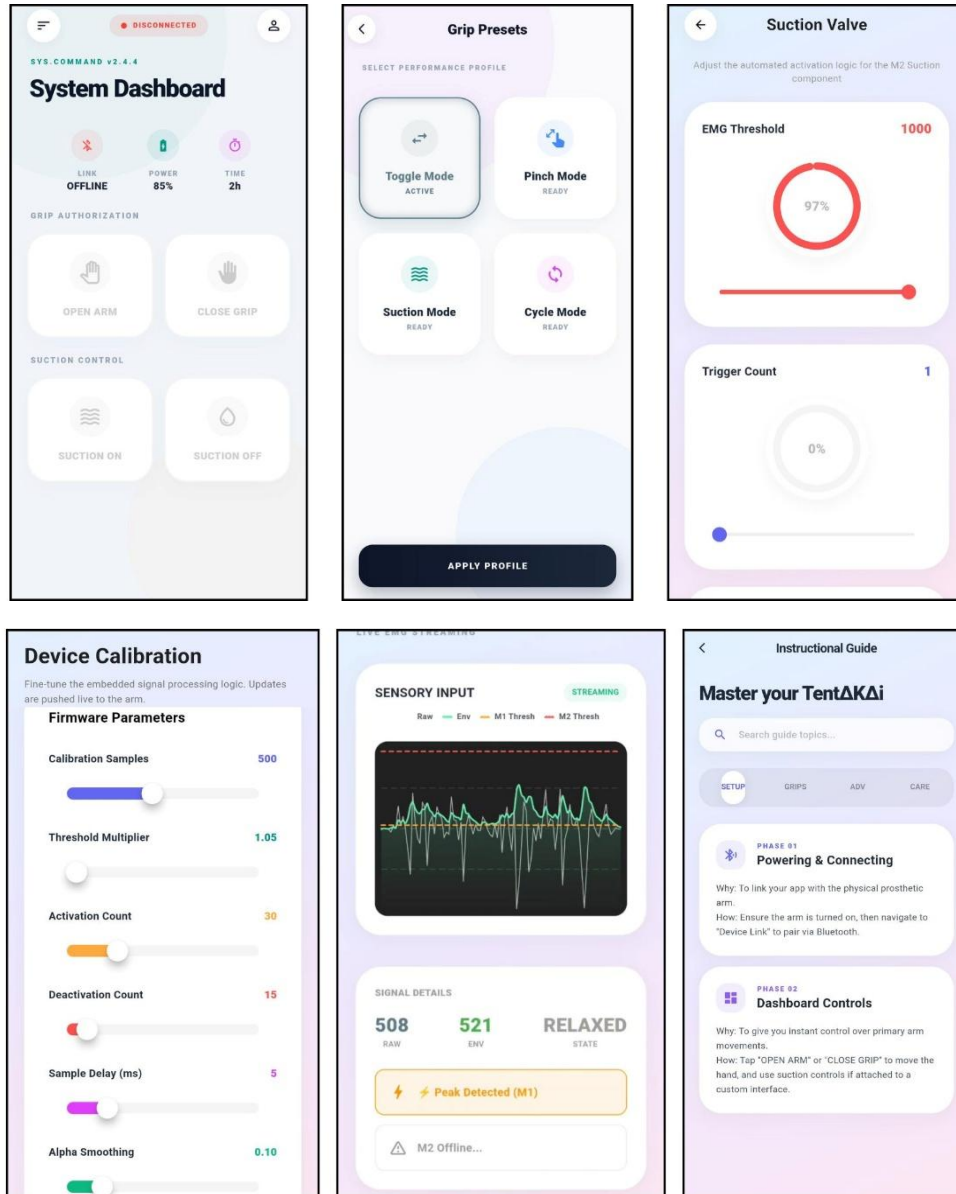


Figure 6.1 Mobile Application User Interface and Functional Modules

As shown in this figure, the user interface is designed for clarity and usability, and provides the following core features:

- **Sensitivity Adjustment:** A sensitivity adjustment slider that can adjust the sensitivity in real time. If the arm is too sensitive while moving normally,

the slider can be adjusted to detect a more deliberate muscle movement.. If the user is fatigued and generating weaker signals, the slider brings the threshold down to ensure the arm remains responsive. This adjustment transmits as a Bluetooth parameter command (P2 threshold multiplier) to the ESP32.

- **Suction Control Override:** Although it is possible to program the vacuum motor to operate simultaneously with the main gripper, the app offers a special control override module panel. This allows the operator to activate or deactivate the vacuum motor manually or set its strength relative to the object that is being grasped. The letter 'L' commands the motor to run in the forward direction, 'R' in reverse direction, and 'S' shuts down the motor.
- **Preset Grip Modes:** The application features multiple pre-programmed modes for specific daily tasks. These include holding a cylindrical cup, picking up a pen, and lifting a flat object, among others. These modes alter the values of threshold multiplier, activation counter, and deactivation counter simultaneously, providing an optimal setting for each action with just one touch
- **Real-Time EMG Waveform:** A live scrolling waveform display shows the user their raw EMG signal alongside the processed envelope and the current activation threshold. This gives the user immediate feedback about whether their arm is detecting their muscle contractions correctly, and helps them understand how to position the socket for the best signal quality.
- **Integrated User Manual:** The integrated digital guide inside the program includes instructions on setting up, operational safety, troubleshooting procedures for common problems, and how each mode works. This ensures that users are not dependent on any outside help, allowing them to solve small issues on their own.

6.4. OPERATING MODES

The embedded program running on the ESP32 supports four distinct operating modes, selectable through the two physical tactile switches (SW1 and SW2) on the socket:

Mode 1 (SW1 OFF, SW2 OFF) — Normal Toggle: Each EMG activation event toggles the gripper between open and closed states. This is the standard daily-use mode for most grasping tasks.

Mode 2 (SW1 ON, SW2 OFF) — Forward Only: Each EMG activation drives the gripper in the closing direction only, for a duration proportional to how long the activation persists. Useful for tasks requiring precise grip depth control.

Mode 3 (SW1 ON, SW2 ON) — Suction Mode: EMG activation controls the suction motor rather than the mechanical gripper. The main gripper is held stationary. Used when handling flat or smooth objects.

Mode 4 (SW1 OFF, SW2 ON) — Cycle Mode: Each EMG activation steps the gripper through a repeating sequence — forward, stop, backward, stop — and cycles again on the next activation. Provides an alternative grip modality for users who prefer not to hold a sustained contraction.

6.5. SUMMARY

The software includes an adaptive EMG calibration algorithm along with an intuitive mobile application known as TentaKai that guarantees accurate and personalized control of the artificial arm. Due to the constant adjustment according to unique muscle movements, the performance level of the software does not change, even if there are changes such as fatigue or sensors' positions. The mobile application allows adjusting sensitivity levels, suction force, pre-defined grip types, and live monitoring of EMG activity, whereas the ESP32 microcontroller provides several operational modes for various purposes.

7. DESIGN FEATURES

7.1. MECHANICAL DESIGN FEATURES

The departure from a traditional five-finger anthropomorphic prosthetic design was not simply a cost-cutting measure it produced genuine mechanical advantages that improve the device's practical utility as a wearable assistive technology.

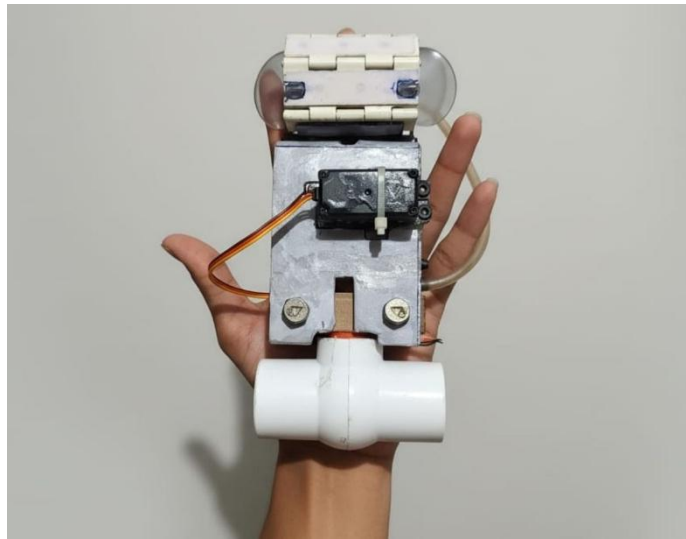


Figure 7.1 Reduced-Bulk Design

- Reduced Bulkiness and Improved Aesthetics:** As a result of condensing the entire process of multi-finger operation into one continuous adaptive component, the size of the overall prosthesis becomes smaller and, thus, the overall volume is reduced. On the one hand, this will reduce the number of parts used, making the construction simpler, and on the other hand, this will improve the usability of the prosthesis. Reduction in the mass leads to decreasing the moment arm about the socket joint, resulting in reducing the weight of the prosthesis and its comfortable usage. People will find it easier to wear the prosthesis for longer periods without experiencing excessive fatigue. Not only will this improvement be useful from a functional point of view, but it will also look more natural for the user.

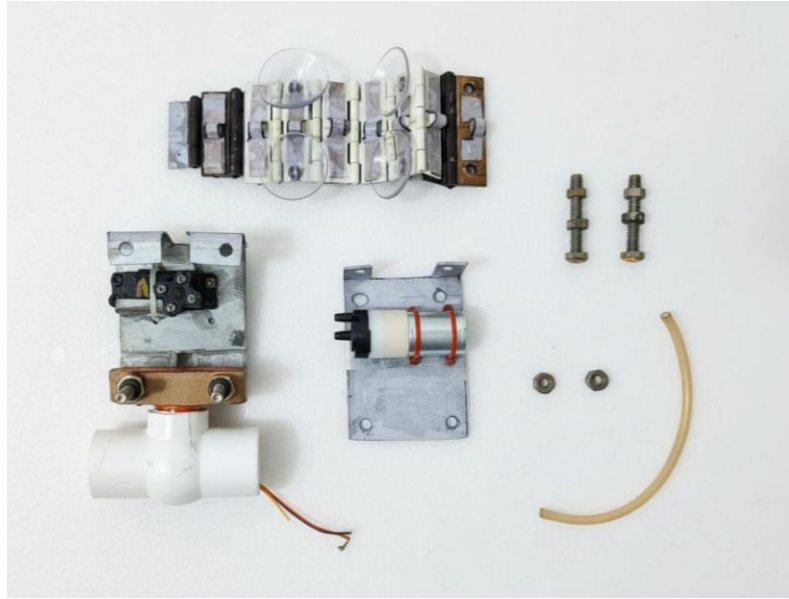


Figure 7.2 Modular System Design

- **Modular Design Approach:** The electrical bay housing the ESP32, motor driver, and power management components is a separate, removable module from the mechanical actuator assembly. In other words, even if the servo motor gets damaged, it would not be necessary to tamper with the electronic components when replacing it with another servo motor. Alternatively, if the microcontroller requires repair or programming, no changes will be made to the mechanical system. This degree of modularity is unusual in prosthetics design and solves the identified issue of repairability.

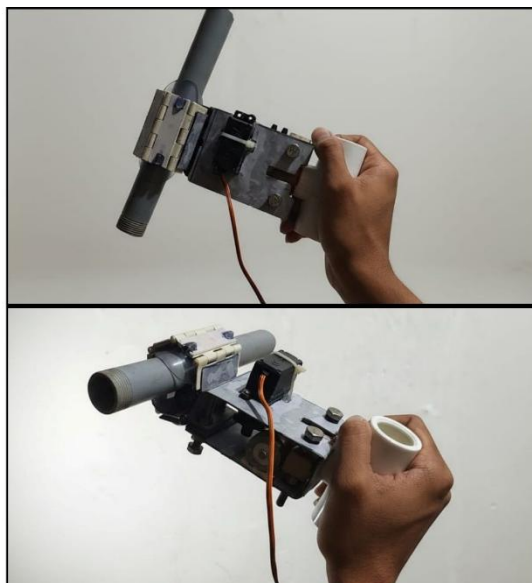


Figure 7.3 Passive Wrist Rotation Enabling Flexible Object Orientation

- **Wrist Rotation Feature:** A passive wrist rotation mechanism allows the terminal device to be rotated to any desired orientation before gripping. This seemingly simple feature significantly expands the utility of the arm. By allowing the user to align the gripper with the natural orientation of an object, it reduces the need for awkward or exaggerated body movements. Instead of repositioning the entire arm, shoulder, or torso to achieve a proper grip, the user can make quick, intuitive adjustments at the wrist. This not only makes everyday tasks more efficient but also enhances comfort and reduces physical strain, resulting in a more natural and practical interaction with the environment.

7.2. SUMMARY

The mechanical design has been optimized through simplicity, comfort, and ease of use to make sure that this prosthetic is useful on an everyday basis. By substituting a conventional multi-finger hand with a simplified, adaptive gripper, the device becomes smaller, more streamlined, and therefore more convenient to control. Such an adjustment not only leads to greater comfort, especially when worn for longer periods of time, but also results in a better aesthetic appearance of the prosthetic. Another significant aspect of the design includes its modularity, which allows easy maintenance by facilitating replacement of individual parts as needed. Also, a simple wrist rotation mechanism makes it possible to adjust the angle of the gripper easily to suit different positions of interaction between the device and objects.

8. AUTOMATING THE DESIGNING OF SOCKET

8.1. THE HIDDEN COST OF CONVENTIONAL PROSTHETIC SOCKETS

When discussing prosthetic arms, commercial pricing often focuses only on the terminal device the gripper or bionic hand itself. However, this represents only part of the total cost. A prosthetic arm cannot be used without a properly fitted socket, which serves as the critical mechanical and biological interface between the amputee's residual limb and the prosthesis.

Socket development in conventional practices involves extensive efforts as it is one of the most specialized tasks of prosthetics. Here, the professional would need to measure the limb manually, develop a model for the socket and then fabricate it. Therefore, the fabrication of a customized socket is both time-consuming and involves a great deal of skill.

The cost of such a customized socket usually ranges between ₹20,000 and ₹50,000 and at times even exceeds this range. It should be noted that the price does not only depend on the material used for the socket but is also based on labor and design efforts.

8.2. MOTIVATION FOR AUTOMATION

Recognizing this challenge, our project extends beyond the prosthetic arm itself. We developed an integrated web-based platform called **ProSocket 3D**, specifically designed to automate the traditionally expensive and time-consuming socket design process. The central idea is simple yet transformative:

Instead of requiring a trained designer to manually create every socket, the software generates a fully customized socket automatically from patient measurements.

This approach dramatically reduces:

- Design time
- Clinical dependency

- Manufacturing complexity
- Overall cost

By shifting socket design from a specialist-driven workflow to a software-driven workflow, the prosthetic becomes truly accessible.

8.3. APPLICATION OVERVIEW

ProSocket 3D is a browser-based parametric CAD application that runs entirely on the client device, requiring no external server or expensive software licenses.

The application performs four essential functions:

- Generates anatomically accurate socket geometry.
- Allows advanced structural customization.
- Integrates housing for embedded electronics.
- Exports ready-to-print manufacturing files.

This transforms what traditionally takes several days into a process that can be completed in minutes.

8.4. THREE-STEP DESIGN WORKFLOW

Step 1: Patient Measurement Acquisition

The clinician begins by entering the patient's anatomical information.

This includes:

- Amputation type (Transradial, Wrist Disarticulation, Transhumeral, or Elbow Disarticulation)
- Left or right limb selection
- Residual limb length
- Terminal stump shape
- Multiple diameter measurements along the stump length

These cross-sectional measurements define the internal contour of the socket. The software uses them to reconstruct an accurate three-dimensional representation of the patient's limb.

Figure 8.1 Measurement Input Interface for Customized Prosthetic Design

Step 2: Interactive 3D Socket Generation

Once measurements are entered, the software instantly generates a fully parametric 3D socket model.

The user can rotate, zoom, and inspect the design in real time while adjusting key structural parameters such as:

- Wall thickness
- Liner clearance
- Trim line height
- Proximal flare
- Overall scaling

Additional optional enhancements include:

- Reinforcement ribs for increased structural strength
- Ventilation holes for thermal comfort
- Integrated electronics bay for housing the ESP32, motor driver, and battery system
- Adjustable bay position, dimensions, and orientation

This level of control allows each socket to be optimized for both comfort and functionality.

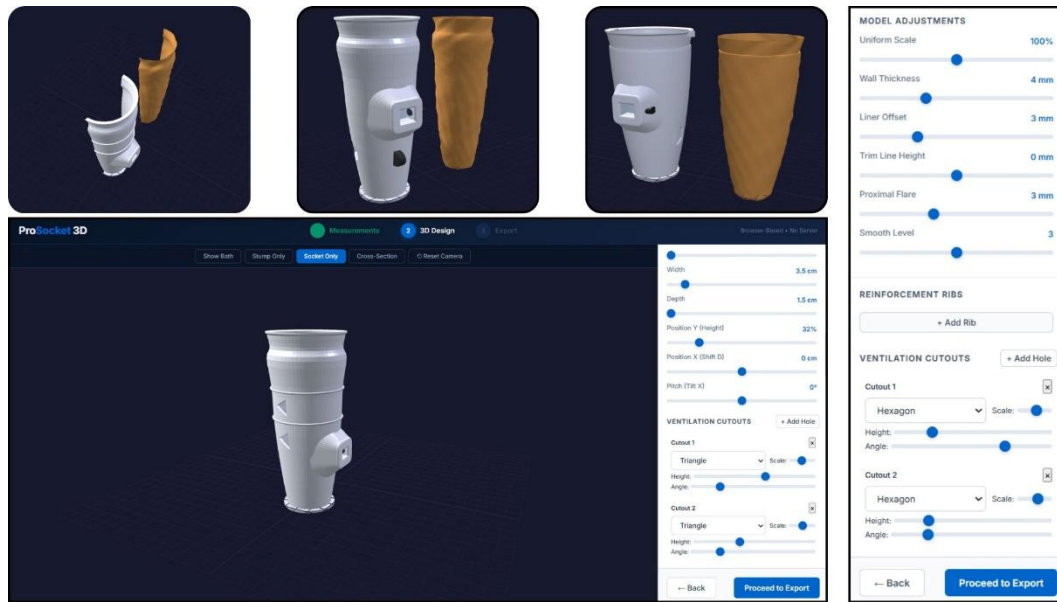


Figure 8.2 3D Socket Design & Parameter Customization

Step 3: Manufacturing and Export

After final validation, the application automatically prepares the socket for fabrication.

The finalized model can then be exported in industry-standard formats:

- STL for direct slicing and 3D printing
- OBJ for advanced CAD workflows
- TXT for complete patient measurement documentation

This allows immediate transition from design to manufacturing.

8.5. SUMMARY

The ProSocket 3D is a browser-based software which designs personalized prosthetic socket automatically based on the measurements from the stump of the patient. The system produces a ready to print model of a prosthetic socket which incorporates features like thickness adjustment, venting, reinforcing ribs, and an electronics bay. This helps save the manufacturing cost and time considerably by obviating the need for designing the socket manually.

9. BUSINESS PERSPECTIVE & COST COMPARISON

9.1. ECONOMIC NEED FOR AFFORDABLE PROSTHETICS

The greatest barrier preventing widespread adoption of advanced upper-limb prosthetics is not technological it is economic. While modern bionic arms offer impressive functionality, their high purchase cost places them far beyond the reach of most amputees, particularly in developing countries and low-income communities. For many families, the price of a commercial prosthetic can exceed several months or even years of household income.

This cost barrier creates a troubling reality: highly capable prosthetic technology exists, yet remains inaccessible to the very people who need it most. As a result, many amputees either continue without any prosthesis at all or rely on simple body-powered devices that offer limited functionality and poor user satisfaction.

9.2. COST STRUCTURE OF CONVENTIONAL PROSTHETIC ARMS

Traditional myoelectric prosthetic arms are expensive due to their complex design and specialized manufacturing. Major factors contributing to their high cost include:

- Multiple precision motors for individual finger control.
- Advanced control electronics and proprietary software.
- Specialized clinical fitting and calibration procedures.
- Additional custom socket fabrication, which is often sold separately and significantly increases the total cost.

These factors make advanced prosthetic arms financially inaccessible for many users.

9.3. COST REDUCTION THROUGH DESIGN SIMPLIFICATION

The prosthetic arm was designed to minimize cost without compromising functionality. Key cost-saving features include:

- A single continuous servo replaces multiple finger motors.
- Simple string actuation eliminates complex tendon mechanisms.
- A low-cost DC motor enables suction-assisted gripping.
- Standard, off-the-shelf electronic components are used throughout.
- The structure is optimized for local manufacturing using 3D printing.
- The custom socket cost is further reduced using the **ProSocket 3D** web application, which automates socket design and generates a ready-to-print 3D model, eliminating expensive manual design and fitting processes.

These design choices significantly lower manufacturing, socket fabrication, repair, and maintenance costs.

9.4. COMPARATIVE ANALYSIS

A direct comparison clearly illustrates the economic advantage of the proposed system. As shown in Table 9.1, the proposed prosthetic arm differs significantly from conventional alternatives across multiple dimensions including motor count, mechanical complexity, EMG control, maintenance burden, and overall cost.

Table 9.1 Comparison of conventional prosthetic arms and proposed prosthetic arm

Conventional Prosthetic Arms	Proposed Prosthetic Arm
Uses multiple motors (5 or more for individual finger control)	Uses only two motors (Continuous servo for gripping and DC motor for suction)
Complex tendon, tendon, gear and finger mechanisms	Simple single string-based gripping mechanism
Advanced multi-channel EMG control systems	Simplifies single-channel EMG control
High manufacturing, repair and maintenance cost	Easy to repair, off-shelf parts, low maintenance
Approximate Cost: 1 Lakh	Estimated cost: 12,000-15,000

9.5. DETAILED COST BREAKDOWN

The project was developed entirely using commercially available components, ensuring easy procurement and future scalability. Table 9.2 provides a detailed breakdown of each component and its approximate cost in INR. The total estimated cost comes to ₹11,100, which confirms that the system can be built at a fraction of the cost of conventional prosthetic arms.

Table 9.2 Detailed Cost Breakdown

Component	Approximate Cost (INR)
Continuous Servo Motor (150 Kg)	3000
Suction Motor & Suction Cup	500
EMG Sensor	1700
Motor Driver & Buck Converters	900
Microcontroller (ESP32) & Switches	1000
Power Supply (3S Li-ion pack)	2000
Miscellaneous (Wiring, 3D printing, Hardware)	2000
Total Estimated Cost	11100

The total system cost remains a fraction of conventional commercial alternatives while still delivering meaningful functionality.

9.6. MAINTENANCE AS A BUSINESS ADVANTAGE

Affordability extends beyond the initial purchase price. Many advanced prosthetic systems become financially burdensome because of expensive repairs and proprietary replacement parts. The low component costs reflected in Table 9.2 further underscore this advantage — in contrast, the proposed design uses standard, readily available components that can be replaced locally without specialized tools or manufacturer intervention. This dramatically lowers the total cost of ownership over the device's lifetime.

9.7. MARKET IMPACT

The target market for this device is enormous. Many of upper-limb amputees worldwide lack access to advanced prosthetic technology due to cost constraints.

By offering:

- Functional EMG control
- Adaptive gripping
- Suction-assisted handling
- Smartphone-based customization
- Modular repairability

at an affordable price point, this project addresses a major unmet need in the assistive technology market.

9.8. SUMMARY

The proposed prosthetic arm addresses one of the biggest challenges in modern prosthetics: affordability. By simplifying the mechanical design, using readily available components, and enabling low-cost 3D printing, the system significantly reduces manufacturing and maintenance costs while retaining essential functionality. Its modular construction further lowers long-term ownership costs by allowing easy local repairs. With features such as EMG control, adaptive gripping, suction assistance, and smartphone customization, this design offers a practical and accessible solution for a large population of upper-limb amputees who cannot afford conventional bionic arms.

10. LIMITATIONS AND FUTURE SCOPE

10.1. CURRENT LIMITATIONS

While the proposed prosthetic arm achieves its primary goal of affordability, simplicity, and functional usability, it is important to acknowledge its present limitations.

- The octopus-inspired gripper, although highly adaptable, does not replicate the fine dexterity of individual human fingers. Tasks requiring precise finger manipulation such as typing, buttoning a shirt, or handling very small objects remain challenging.
- The control system currently relies on a single-channel EMG input. This greatly simplifies operation and reduces cost, but it also limits the number of independent commands that can be generated simultaneously. As a result, the range of complex movements is lower than that of premium multi-channel prosthetic systems.
- The suction mechanism is highly effective for smooth and flat surfaces, but its performance depends on the quality of the seal. Rough, porous, or irregular surfaces may reduce suction reliability.
- Finally, although ProSocket 3D significantly reduces socket design cost and time, final clinical fitting and user comfort still require professional supervision. No software can completely replace the expertise of an experienced prosthetist.

10.2. FUTURE SCOPE

The current prosthetic arm establishes a strong foundation for affordable, intelligent, and user-friendly assistive technology. However, its true potential extends far beyond the present design. Several exciting advancements can further improve its functionality, versatility, and real-world impact.

10.2.1. EXPANSION TO HIGHER-LEVEL AMPUTATIONS

The present design is primarily intended for transradial amputees. Future versions can be extended to support elbow-level and shoulder-level amputations by incorporating additional powered joints and control mechanisms. This would significantly broaden the range of users who could benefit from the system.

- Extension to elbow-level prosthetic systems
- Extension to shoulder-level prosthetic systems

10.2.2. ADVANCED EMG CONTROL

While the current single-channel EMG system offers simplicity and reliability, future designs can leverage more sophisticated signal-processing algorithms and machine learning techniques. This would improve gesture recognition, reduce false activations, and allow more natural control.

- Improved EMG signal processing techniques
- AI-based adaptive EMG learning for enhanced control accuracy

10.2.3. SENSORY FEEDBACK

A major limitation of most low-cost prosthetics is the lack of sensory feedback. Integrating tactile and force sensing would allow users to better understand how firmly they are holding an object, reducing accidental drops or excessive gripping force.

- Addition of force or tactile feedback
- Integration of force and pressure sensors for real-time grip monitoring

10.2.4. INDUSTRIAL APPLICATIONS

The octopus-inspired adaptive gripping mechanism has applications beyond prosthetics. Its ability to conform to irregular shapes makes it highly suitable for industrial automation, warehouse handling, and robotic manipulation.

- Adaptation for industrial robotic arms
- Deployment in factory automation and smart manufacturing systems

10.3. SUMMARY

The current prosthetic arm successfully balances affordability, simplicity, and functionality, though it has some limitations in fine finger dexterity, multi-command control, and suction performance on irregular surfaces. Despite these constraints, it provides a strong foundation for future development. Upcoming improvements could include support for elbow- and shoulder-level amputations, advanced AI-based EMG control, integration of tactile and force feedback, and adaptation of the gripping mechanism for industrial robotics and automation. These advancements have the potential to greatly expand both its capabilities and its real-world applications.

11. CONCLUSIONS

The development of this EMG-controlled prosthetic arm was driven by a simple but important goal: to make advanced prosthetic technology accessible to people who need it most. Existing upper-limb prostheses often force users to choose between affordability and functionality. High-end bionic arms offer excellent performance, but their cost, complexity, and maintenance requirements place them beyond the reach of many amputees.

This project demonstrates that a different approach is possible.

By drawing inspiration from the adaptive gripping ability of an octopus tentacle, the conventional multi-finger design was replaced with a simpler, stronger, and more versatile gripping mechanism. Using only two motors, a single-channel EMG interface, and an integrated suction system, the prosthetic delivers reliable everyday functionality while drastically reducing mechanical complexity.

The addition of the TentaKai mobile application further enhances usability by allowing users to calibrate muscle sensitivity, control suction, and customize grip settings according to their daily needs. This ensures that the prosthesis adapts to the user, rather than forcing the user to adapt to the prosthesis.

Equally important is the introduction of ProSocket 3D, a web-based platform that automates the design of custom prosthetic sockets. Since socket fabrication often represents a significant portion of total prosthetic cost, automating this process dramatically improves affordability and accessibility while reducing clinical dependency.

Together, these innovations create a prosthetic system that is:

- Affordable
- Reliable
- Easy to operate
- Easy to maintain
- Suitable for real-world daily use

The project successfully bridges the gap between low-cost prosthetics and advanced bionic functionality.

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ANNEXURE

APP CODE

```
void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  SystemChrome.setPreferredOrientations([
    DeviceOrientation.portraitUp,
    DeviceOrientation.portraitDown,
  ]);
  runApp(const TentaKaiApp());
}

class TentaKaiApp extends StatelessWidget {
  );
}
}
```

The Command Protocol Layer ('command_service.dart')

This service acts as an abstraction layer over the raw Bluetooth service. It enc

```
class CommandService {
  final BluetoothInterface _bluetoothService;
  CommandService(this._bluetoothService);

  // Grip control
  void gripOpen() =>
    _bluetoothService.sendCommand(AppConstants.cmdGripOpen);
  void gripClose() =>
    _bluetoothService.sendCommand(AppConstants.cmdGripClose);
  // Suction control
  void suctionOn() =>
    _bluetoothService.sendCommand(AppConstants.cmdSuctionOn);
```

```

void suctionOff() =>
  _bluetoothService.sendCommand(AppConstants.cmdSuctionOff)
void setSuctionLevel(int level) =>
  _bluetoothService.sendCommand(AppConstants

// Emergency Stop
void emergencyStop() =>
  _bluetoothService.sendCommand(AppConstants.cmdStopAll)

// Preset Modes
void setModeLight() =>
  _bluetoothService.sendCommand(AppConstants.cmdModeLight
void setModeBottle() =>
  _bluetoothService.sendCommand(AppConstants.cmdModeBott)
}

```

Real-time Waveform Rendering ('mini_waveform.dart')

To visualize high-frequency EMG data without performance bottlenecks, the app
us

```

class _MiniWavePainter extends CustomPainter {
  final List<double> points;
  final int maxPoints;
  final Color color;

  _MiniWavePainter({required this.points, required this.maxPoints, required this

  @override
  void paint(Canvas canvas, Size size) {
    if (points.isEmpty) return;

    final double width = size.width;
    final double step = width / (maxPoints - 1);

```

```
final Path path = Path();
path.moveTo(0, size.height - (points[0] / 80 * size.height).clamp(0.0, size
```

```
for (int i = 1; i < points.length; i++) {
final double x = i * step;
final double y = size.height - (points[i] / 80 * size.height).clamp(0.0
path.lineTo(x, y);
}
```

```
final Paint linePaint = Paint()
..color = color.withOpacity(0.8)
..style = PaintingStyle.stroke
..strokeWidth = 1.5
..strokeCap = StrokeCap.round;
```

```
canvas.drawPath(path, linePaint);
}
```

```
@override
bool shouldRepaint(covariant _MiniWavePainter oldDelegate) => true;
}
```

```
const TentaKaiApp({Key? key}) : super(key: key);
```

```
@override
Widget build(BuildContext context) {
return MultiProvider(
providers: [
ChangeNotifierProvider<BluetoothInterface>(create: (_) => getBluetoothSe
ChangeNotifierProvider<SettingsProvider>(create: (_) => SettingsProvider],
child: Consumer<SettingsProvider>(
builder: (context, settings, child) {
```

```

return MaterialApp(
  title: 'TentKai',

  debugShowCheckedModeBanner: false,
  theme: AppTheme.getTheme(isHighContrast: settings.isHighContrast),
  home: const SplashScreen(),
);
},
),

```

PROGRAM TO RUN THE ARM

- **Pin Definitions & Tunable Parameters**

```

#include <SoftwareSerial.h>

SoftwareSerial BT(2, 3);

const int EMG_PIN    = A0;
const int SWITCH_PIN  = 7;
const int SWITCH_FWD_PIN = 6;
const int IN1 = 8, IN2 = 9;
const int IN3 = 10, IN4 = 11;

int  calibrationSamples = 500;
float thresholdMultiplier = 1.05;
int  activationCount    = 30;
int  deactivationCount  = 15;
int  sampleDelayMs      = 5;
float alpha              = 0.10;

int  m2RawThreshold = 1000;
int  m2TriggerCount = 1;
int  m2ResetCount   = 20;

```

- **Calibration**

```
void calibrate() {  
    long sum = 0;  
    for (int i = 0; i < calibrationSamples; i++) {  
        sum += analogRead(EMG_PIN);  
        delay(sampleDelayMs);  
    }  
    baselineEMG = sum / calibrationSamples;  
    threshold = (int)(baselineEMG * thresholdMultiplier);  
}
```

- **EMG Envelope Detection & Threshold**

```
rawEMG = analogRead(EMG_PIN);  
  
float rectified = abs(rawEMG - baselineEMG);  
envelope = (int)(alpha * rectified + (1.0 - alpha) * envelope);  
bool muscleActive = (envelope > (threshold - baselineEMG));  
  
if (muscleActive) {  
    aboveCount++;  
    belowCount = 0;  
} else {  
    belowCount++;  
    aboveCount = 0;  
}
```

Mini project report new.pdf

 Mar Baselios College of Engineering and Technology

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